

States, Input and Loops

LEARNING OBJECTIVES



You build a scene with multiple game objects and learn about the **Game Loop**, **Input** and **Signals** as core systems of a game.

01 Build the Nodes ◦

Use the scene from Assignment 1 or create a new scene with `Node2D` as the root node. Add at least three `Area2D` nodes as children. Each `Area2D` gets a `Sprite2D` with the Godot logo (`icon.svg`) and a `CollisionShape2D` as children. Experiment with properties like `Scale`, `Modulate` and `Rotation` in the Inspector.

02 Game Loop ◊

Attach a script to the root node. Implement `_process(delta)` and use it to continuously rotate all child nodes.

`delta` ensures framerate-independent animation by accounting for the time elapsed between frames.

03 Input ▷

Define three custom actions in the **Input Map**, one per `Area2D`. When the assigned key is pressed, set the `modulate` of the corresponding `Area2D` to a new color.

The Input Map is found under **Project** → **Project Settings** → **Input Map**. You can assign custom action names and bind them to any key.

04 Signals & Click Interaction ★

Attach a script to each `Area2D` and connect its `input_event` signal. A click should produce the same effect as the corresponding key press from Task 03.

For clicks to be detected, `Input Pickable` must be enabled on the `Area2D`. Check the event type using `event is InputEventMouseButton`.

RESOURCES

- [Godot Docs: Idle & Physics Processing](#)
- [Godot Docs: Input Events](#)
- [Godot Docs: Signals](#)

ACCEPTANCE REQUIREMENTS

- ☐ At least three visible objects are arranged in the scene.
- ☐ A running loop visibly changes a property of the objects.
- ☐ Three distinct inputs are defined; each visibly changes a property of a corresponding object.
- ☐ Clicking an object visibly changes at least one property of the clicked object.